



**MSC ARTIFICIAL INTELLIGENCE
DEEP LEARNING AND GENERATIVE AI**

SUBMITTED BY

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Introduction

Informal settlements continue to be a major problem in many low- and middle-income regions rapid urbanization occurs much faster than it does in terms of planned development and related infrastructure. The need to identify the locations of slums accurately is critical for local governments, international humanitarian agencies, planners of public works, and those involved in sustainable development efforts. Mapping slums is a laborious process and is generally conducted by human mappers who conduct assessments over large geographic areas. Because of this, remote sensing technologies (satellite imagery) and deep learning techniques have been gaining popularity as methods for automatically detecting slums. These two technologies offer advantages in terms of being able to cover larger geographic areas more quickly than human mappers.

Recently, studies have demonstrated that deep learning networks can extract relevant features in remote sensing data, including: density, textures, arrangements of buildings, and surface properties associated with informal settlements. Specifically, among recent developments in deep learning networks, the Mixture-of-Experts (MoE) architecture holds promise for identifying slums because each MoE sub-network can focus on extracting specific feature patterns, whereas a common feature extractor captures more general representations. The selected paper, Generalizable Slum Detection from Satellite Imagery with Mixture-of-Experts, proposes such a framework to address the challenge of domain shift in slum detection across regions. In Part 1, this paper was critically appraised and extended through a proposal for an uncertainty-aware Mixture-of-Experts model using Monte Carlo Dropout.

In Part 2 of this project, a working deep learning artefact for detecting slums via satellite images using PyTorch will be implemented. To account for limitations in terms of data availability and resource constraints within the project's scope, the current project will utilise satellite data only over Mumbai instead of adopting a full adaptation approach applicable across multiple regions. Three different versions of the artefact will be evaluated in this project: a baseline version that utilises a Convolutional Neural Network (CNN), a version utilizing an Mixture-of-Experts (MoE) model and a version utilising both an MoE model and Monte Carlo dropout at inference.

The aim is to evaluate whether the expert-based architecture and uncertainty estimation can improve slum classification performance and provide more reliable predictions in ambiguous cases.

Background and Related Work

Automated detection of slums via satellite images has gained attention among researchers due to the costliness, difficulty of updating, and time required for traditional field surveys. According to a recent meta-analysis of slum mapping studies conducted since 2014 and up until 2024, the combination of remote sensing and deep learning is currently being utilised within this area; however, the meta-analysis indicated that no universal method exists to create a generalised model that will capture all variations of slum characteristics across geographically disparate areas. Therefore, generalizability is one of the primary challenges facing researchers in this area [1].

In addition to addressing the issue of generalizability, another significant trend in remote sensing is the development of convolutional neural networks (CNNs). The reason why CNNs are so successful at extracting hierarchical spatial features (i.e., edges, texture, local shape, neighbourhood pattern) is that they do so from raw image data. In fact, a comprehensive review of deep learning for remote sensing image classification reported that CNN-based methodologies are now considered to be commonplace for both scene-level and pixel-level classification tasks [2]. This may help explain why many researchers utilise CNNs as a strong baseline before evaluating other potentially more complex architectures.

Researchers have also attempted to apply transfer learning to address issues associated with slum mapping. For example, Verma et al. demonstrated that transfer learning can aid in the identification of urban slums using pre-trained convolutional networks along with high-resolution satellite imagery [3]. This is particularly important because remote sensing projects typically face challenges regarding the availability of labelled ground truth data, especially in resource-poor settings.

Another line of research addresses the application of segmentation models versus patch-level classification. Architectures such as U-Net are specifically influential due to the ability of U-Nets to retain fine-grained spatial detail while also providing sufficient contextual information in order to accurately perform pixel-level region boundary delineation [4]. Although U-Net was originally developed for biomedical image segmentation, it has been extensively adopted as a generic segmentation architecture, including those applied in remote sensing workflows, where accurate spatial boundaries are critical.

This selected paper expands upon existing research by introducing GRAM (Generalised Region-Aware Mixture-of-Experts) for detecting slums. Unlike previous methods that rely upon a single decision pathway, this paper utilises a Mixture-of-Experts (MoE) design wherein multiple expert sub-networks specialise in identifying various types of feature patterns and the output of each sub-network is combined by a gating mechanism. The authors contend that this MoE design is beneficial because slum morphologies vary significantly across cities and regions, thus resulting in a considerable degree of domain shift [5]. As such, the authors combine expert specialisation with adaptation mechanisms to improve performance on unseen locations and report results on a large-scale dataset that spans twelve cities across four continents.

Additionally, recent research has emphasised estimating uncertainty. Since standard neural networks can produce erroneous classifications with high confidence levels, this is detrimental in socially sensitive applications such as slum mapping. Gal and Ghahramani showed that dropout could be interpreted as approximating Bayesian inference and that Monte Carlo Dropout could be applied at test-time to estimate predictive uncertainty through repeated stochastic forward passes. This represents a practical method to evaluate the reliability of a prediction without employing a fully Bayesian neural network [6]. As such, this is advantageous for high-stakes urban mapping tasks, as it allows researchers to differentiate certain classifications made with confidence from more ambiguous classifications.

Collectively, the literature illustrates that researchers have employed three primary approaches to address slum mapping and similar remote sensing problems: CNN-based classification, transfer-learning and segmentation models for spatial mapping, and more recently expert-based or uncertainty-aware methodologies to account for heterogeneity and predictability. Furthermore, the existing research establishes the basis for this study's utilisation of a CNN baseline model, a Mixture-of-Experts model and an uncertainty-aware extension model.

Methodology

In this project, a practical deep learning pipeline for slum classification in satellite images was developed. It has been implemented in Python on top of PyTorch, and three different model configurations have been tested: a baseline Convolutional Neural Network (CNN), a Mixture-of-Experts (MoE), and an uncertainty-aware Mixture-of-Experts (UA-MoE) with Monte Carlo Dropout during inference. The paper originally suggested the Generalizable Remote Sensing-

based Adaptation Model (GRAM) for cross-region slum detection, but the data used for this project included imagery from only Mumbai. Therefore, the focus was on replicating the core Mixture-of-Experts architecture and adding uncertainty estimation to it, instead of implementing the entire domain adaptation framework.

Dataset and Patch Preparation

The data comprised NumPy arrays of size (4, 3954, 2105) for a satellite image and (3954, 2105) for a binary mask of slums. RGB visualisations were created using 3 channels for inspection and all 4 channels in the model training. The RGB image was compared visually with the mask and overlay to verify the annotations corresponded with the satellite imagery.

The original image was represented by a single large image, so the image patches were created using a sliding window. The stride was 64, and the patch size was 64 to get non-overlapping patches. The binary classes were assigned to each image patch according to the percentage of slum area in its mask. An average mask value greater than 0.01 on a patch was classified as a slum, and the rest as non-slum. This process generated 1,952 image patches with dimensions (1952, 4, 64, 64). While this simplified the original pixel-level segmentation task to patch-level binary classification, it significantly reduced computational complexity while being appropriate for comparative model evaluation.

Data Splitting and Preprocessing

The data was partitioned into training, validation and test sets with stratified sampling to maintain the class proportion. The patches were split into three groups: training set—70% (1,366 patches); validation set—30% (293 patches); and testing set—30% (293 patches). To avoid information leakage, normalisation statistics were computed only with the training data and then applied to the validation and test sets. An efficient mini-batch processor was implemented using custom PyTorch Dataset and DataLoader objects. The labels were used to calculate balanced class weights to minimise any class imbalance in the training set and applied to the cross-entropy loss function.

Model Architectures

The baseline CNN was made up of three convolutional blocks containing a convolutional layer, batch normalisation, ReLU, max-pooling, followed by fully connected layers with dropout, before being sent to the binary classification layer. This model was used for comparison.

The second one was a Mixture-of-Experts architecture based on the same shared convolutional feature extractor, followed by three expert subnetworks and a gating network. The gating network provided the softmax weights, which determined the contribution of each expert, enabling the model to integrate the specialist feature representations rather than a single classifier.

Uncertainty-Aware Extension

Monte Carlo Dropout is applied to the trained MoE during testing to assess the prediction confidence. The expert networks had two dropout layers, and the number of stochastic forward passes for each test sample was 20. The final prediction was based on the average class probabilities from all passes, and the average predictive variance was adopted as the uncertainty score.

The uncertainty thresholds were not chosen at random. Monte Carlo Dropout was initially applied to the validation set, and uncertainty thresholds corresponding to the 50th, 60th, 70th, 80th, 90th and 95th percentile were evaluated. The thresholds were evaluated with prediction coverage, manual review rate, accepted prediction accuracy and F1-score. The 90th percentile threshold was chosen for the study on the independent test set due to its optimal trade-off between prediction accuracy and coverage.

Training and Evaluation

Both models were built with the Adam optimiser, a learning rate of 0.001, a batch size of 16 and max epoch of 25. Early stopping was used with a patience of 5 epochs using validation loss, and the model weights corresponding to the best-performing epochs were kept.

The accuracy, precision, recall, F1 score, classification reports and confusion matrices were used to evaluate the performance of the model. Furthermore, uncertainty analysis was performed between correct and incorrect predictions, and an ablation study was conducted between baseline CNN, standard MoE, UA-MoE and threshold-based UA-MoE. Additionally, the number of trainable parameters, inference time per image, and extra computational burden of Monte Carlo Dropout were reported to evaluate computational efficiency.

Experiments and Results

Experimental Setup

The dataset was split into stratified training, validation, and test sets of 1,366, 293 and 293 image patches, respectively. The baseline CNN and Mixture-of-Experts were both trained with the Adam optimiser and a learning rate of 0.001, with a max epoch of 25 and a batch size of 16. The model with the lowest validation loss was chosen using early stopping with a patience of 5 epochs. The performance of the models was assessed through accuracy, precision, recall, F1 score, confusion matrix and uncertainty analysis.

Baseline CNN Results

The baseline CNN yielded an accuracy of 0.8191, precision of 0.6475, recall of 0.8876 and F1-score of 0.7488 on the independent test set.

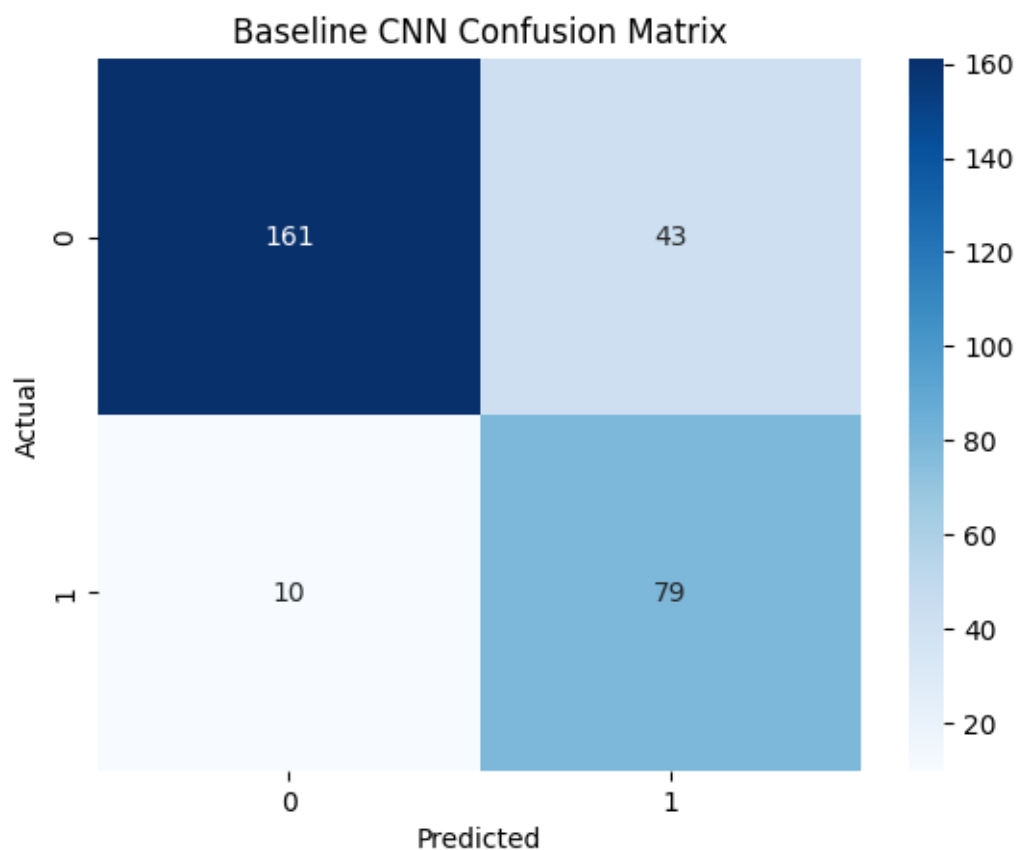


Figure 1. Baseline CNN Confusion Matrices

The confusion matrix illustrates the baseline CNN's accuracy in predicting slum patches with a high recall rate, but also that it misclassified some non-slum patches as slums. This led to a moderate precision value, showing the model preferred to detect slums and so there were more false positives.

Mixture-of-Experts Results

The accuracy, precision, recall, and F1-score obtained by the standard Mixture-of-Experts model were 0.8259, 0.6696, 0.8427 and 0.7463, respectively.

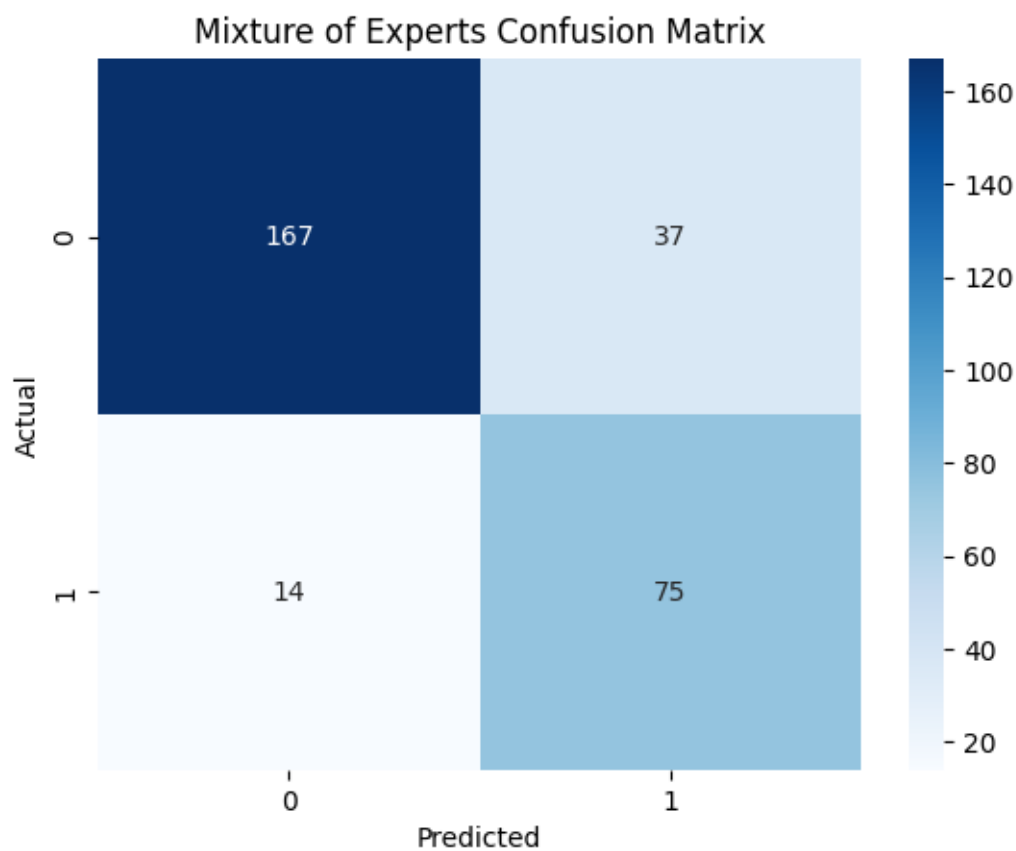


Figure 2. Mixture-of-Experts Confusion Matrices

In overall accuracy and precision, the MoE model outperformed the baseline CNN, and in recall, it was slightly lower than the CNN. The gating network allowed the model to fuse the predictions of the experts, with fewer false positives than the baseline model and with similar general performance.

Uncertainty-Aware MoE Results

During inference, 20 stochastic forward passes were taken with Monte Carlo Dropout. The final prediction was computed by averaging the class probabilities, and predictive variance was an estimate of uncertainty. The accuracy, precision, recall, and F1-score were 0.8191, 0.6607, 0.8315, and 0.7363, respectively, for the uncertainty-aware MoE.

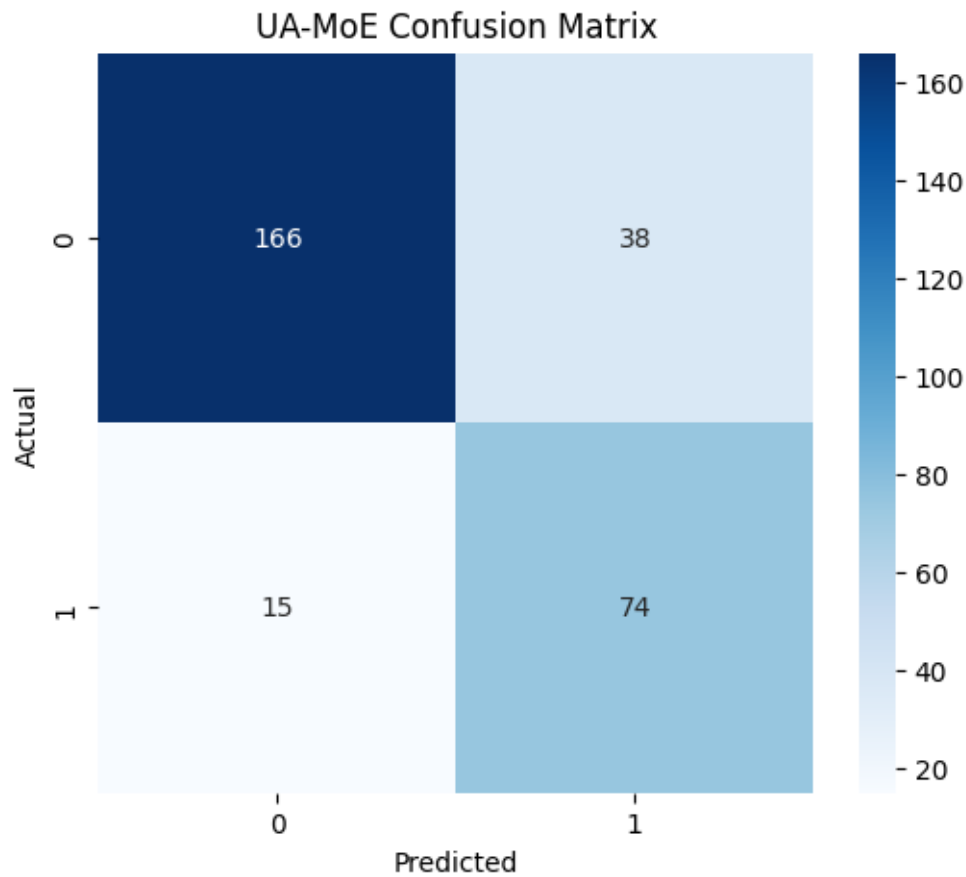


Figure 3. UA-MoE Confusion Matrices

Although Monte Carlo Dropout alone did not improve overall classification performance compared with the standard MoE model, incorrect predictions consistently exhibited higher predictive uncertainty than correct predictions.

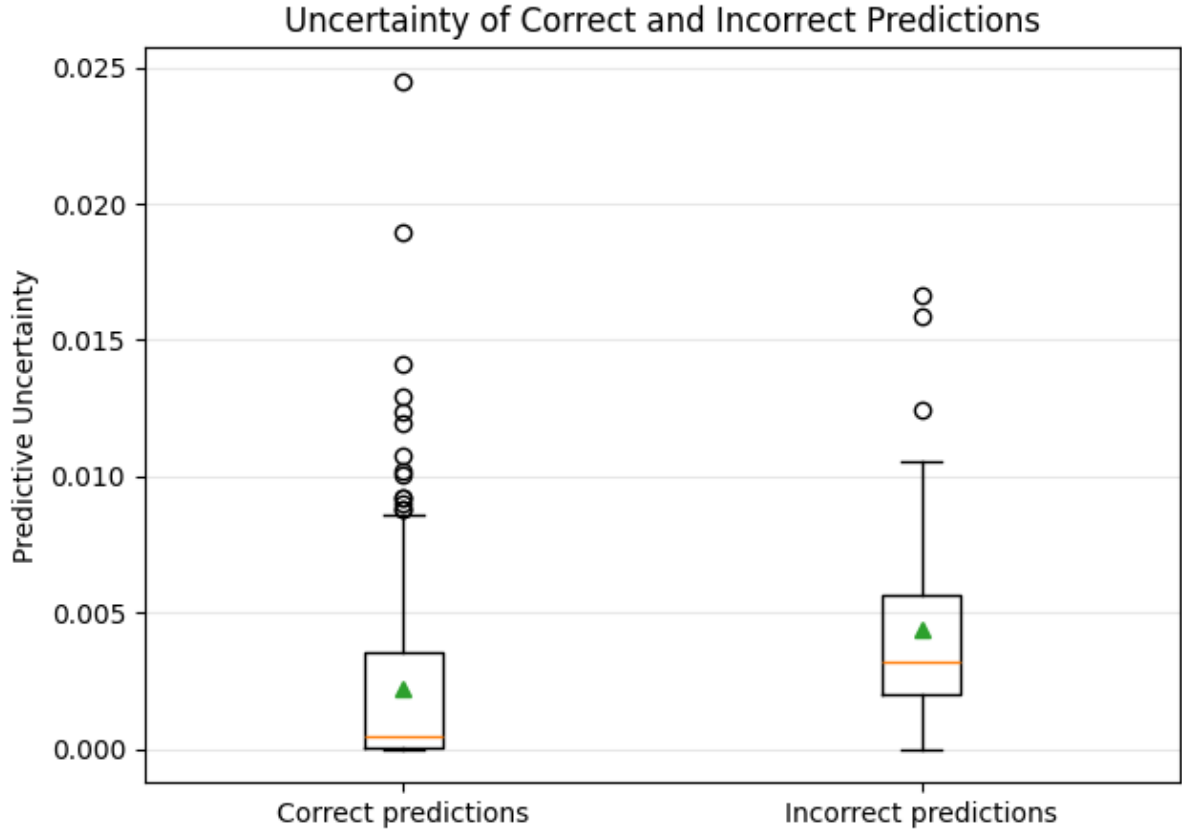


Figure 4. Predictive Uncertainty of Correct and Incorrect Predictions

The boxplot shows that incorrect predictions generally have higher predictive uncertainty than correct predictions. Both the median and mean uncertainty values are larger for incorrect classifications, and the uncertainty distribution is more widely spread. This indicates that the uncertainty-aware MoE model is typically less confident when making incorrect predictions, demonstrating that Monte Carlo Dropout provides meaningful confidence estimates that can be used to identify unreliable predictions for manual review.

Uncertainty Threshold Selection Using Validation

Instead of setting a threshold individually, thresholds for the 50th, 60th, 70th, 80th, 90th and 95th percentiles of the uncertainty distribution for validation were considered.

The 90th percentile was chosen as the best compromise between prediction accuracy and the number of predictions retained, and was used for testing.

The threshold was applied to the test set and was found to have retained 89.42% of the predictions while referring 10.58% of the unclear samples to manual review. The prediction accuracy of the

accepted predictions was 0.8397, and the F1-score was 0.7500, which showed that uncertainty filtering increased the reliability of the predictions and only reduced the number of predictions that were not certain by a small amount.

Computational Performance

The number of trainable parameters and the time for inference with each model were compared to assess the computational complexity.

The baseline CNN had 1.142,946 trainable parameters, and the Mixture-of-Experts had 7.436,841.

The baseline CNN required 0.260 ms per image, whereas the Mixture-of-Experts model required 0.227 ms per image during standard inference. Applying Monte Carlo Dropout increased the total inference time to 0.4384 s for the test set, approximately 12.38 times slower than standard MoE inference because multiple stochastic forward passes were performed.

Ablation Study

An ablation study was done to assess each of the proposed components.

Configuration	Accuracy	Precision	Recall	F1 Score	Coverage
Baseline CNN	0.8191	0.6475	0.8876	0.7488	1.0000
Standard MoE	0.8259	0.6696	0.8427	0.7463	1.0000
MoE + MC Dropout	0.8191	0.6607	0.8315	0.7363	1.0000
MoE + MC Dropout + Uncertainty Threshold	0.8397	0.6632	0.8630	0.7500	0.8942

Table 1. Ablation Study Results

The ablation study indicates that the incorporation of Monte Carlo Dropout alone did not significantly affect the classification performance. The uncertainty estimation with the validation-based threshold selection, however, yielded the highest accuracy with almost 90% of predictions. These findings demonstrate that uncertainty estimation is most valuable as a decision-support mechanism for identifying unreliable predictions for manual review rather than as a method for directly improving classification accuracy.

Discussion

Interpretation of Results

The results show that the patch-based slum classification of satellite images is possible, but still problematic since numerous patterns and textures can be found in both slum and non-slum classes. The basic CNN showed good accuracy (0.8191) and F1 score (0.7488), demonstrating that simple CNN feature extraction was good at learning discriminatory features of images.

The standard Mixture-of-Experts (MoE) model had the highest accuracy (0.8259) and showed a slight improvement in precision over the baseline CNN, but it saw a slight drop in recall. It indicates that the expert and gating architecture lowered false positives without sacrificing overall performance. The uncertainty-aware MoE achieved by using Monte Carlo Dropout did not lead to any further increase in classification accuracy, but the predictive uncertainty of the misclassified predictions was consistently higher than that of the correct predictions. This was verified in the uncertainty boxplot, which showed that for wrong classifications the model had less certainty.

The best improvement was achieved with the uncertainty threshold selection based on validation data. Choosing the 90th percentile threshold kept 89.42% of the predictions, only sending 10.58% of class samples for manual testing. The accepted predictions were successful at 0.8397 and an F1 score of 0.7500, suggesting that the uncertainty information is best used to identify the predictions that are not reliable, rather than directly for increasing the accuracy of the classification.

Challenges and Limitations

One limitation was that the dataset contained satellite imagery from only a single city (Mumbai). This meant the diversity of urban environments was small, which in turn meant there was a limited opportunity to benefit from expert specialisation and to evaluate cross-region generalisation. Moreover, the original segmentation was simplified to a binary classification of patch (slum or non-slum); hence, the spatial complexity was reduced, but the slum boundary was not captured precisely.

Also, adding uncertainty estimation made the computational cost increase significantly. While MoE inference took 0.227ms per image, Monte Carlo Dropout needed around 12.38 times more time due to multiple stochastic forward passes. This is a compromise between the degree of reliability of the prediction and computational efficiency.

Alignment with Project Objectives and Future Work

The project implemented and evaluated the three proposed model configurations, with success: a baseline CNN, a standard Mixture-of-Experts (MoE) model, and an uncertainty-aware MoE model based on Monte Carlo Dropout (MCD). The experimental results showed that the predictive uncertainty can be used to identify less reliable predictions, which is the objective of improving the reliability of decision-making. The GRAM framework, as discussed in the original paper, was not reproduced, as cross-region adaptation and pseudo-label refinement were not possible within the scope of the dataset.

Future work includes the evaluation of the framework with multi-city datasets, exploration of uncertainty-guided gating and pseudo-label filtering, and the application of the method from patch classification to pixel-level semantic segmentation. Other measures of calibration, such as Expected Calibration Error (ECE), can also be taken to assess the quality of the predictions.

Ethics and Scalability

The use of deep learning algorithms for slum detection also poses ethical questions, as prediction inaccuracies could impact disadvantaged communities. The false positives might incorrectly designate formal neighbourhoods as informal settlements, while false negatives could leave a genuinely deprived area out of the planning and/or resource allocation process. So model predictions should be treated not as decisions, but as decision support tools and should be double-checked in cases of uncertainty.

The data from a single city, Mumbai, also causes geographic bias as learned visual patterns may not be generalizable to cities with different settlement characteristics. The uncertainty-aware extension cannot overcome this limitation, but it does give an estimate of prediction confidence, which can be used to flag less reliable predictions for manual review, thereby aiding more responsible decision-making.

The key to scalability is to find a compromise between predictive accuracy and computational efficiency. The baseline CNN had 1,142,946 trainable parameters, while the Mixture of Experts had 7,436,841 parameters. But MC-Dropout took roughly 12.38 times longer to make an inference (0.260 ms/image for CNN and 0.227 ms/image for MoE), due to the need for multiple stochastic

forward passes. Thus, uncertainty estimation can enhance reliability and add extra computational burden, which should be kept in mind when applying the approach to large urban areas.

Conclusion

The following three baseline models were implemented and evaluated in this project: CNN, Mixture-of-Experts and an uncertainty-aware Mixture-of-Experts model for patch-based slum classification using satellite images. Monte Carlo Dropout gave reasonable uncertainty estimates that separated reliable from unreliable predictions, while the standard MoE had the best overall classification performance. The uncertainty threshold selection based on the validation of the model further improved the accuracy of accepted predictions to 0.8397, with 89.42% of the predictions remaining accurate for automatic classification.

The study shows that uncertainty estimation is of utmost value for enabling reliable decision-making instead of directly enhancing classification accuracy. The pipeline was implemented with only one city's dataset available to complete the GRAM framework, and demonstrated its practical application for slum detection, but still provides a basis for future work with larger multi-city datasets and cross-region adaptation.

References

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Appendix:



slum_detection_AI_and
ML_Revise.ipynb